

The Von Neumann Architecture



**Von Neumann
Architecture**



Memory Size/Speed

- Typical memory in a personal computer (PC):
 - 64MB - 256MB – 512MB – 1024MB (1GB) , 1TB, -----
- Memory sizes:
 - Kilobyte (KB) = 2^{10} = 1,024 bytes ~ 1 thousand
 - Megabyte(MB) = 2^{20} = 1,048,576 bytes ~ 1 million
 - Gigabyte(GB) = 2^{30} = 1,073,741,824 bytes ~ 1 billion
- Memory Access Time (read from / write to memory)
 - 50-75 nanoseconds (1 nsec = 0.000000001 sec)
- RAM is
 - volatile (can only store when power is on)
 - relatively expensive



Operations on Memory

- Fetch/Read (address):
 - Fetch a copy of the content of memory cell with the specified address.
 - Non-destructive, copies value in memory cell.
- Store/Write (address, value):
 - Store the specified value into the memory cell specified by address.
 - Destructive, overwrites the previous value of the memory cell.



Input/Output Subsystem

- Handles devices that allow the computer system to:
 - Communicate and interact with the outside world
 - Screen, keyboard, printer, ...
 - Store information (mass-storage)
 - Hard-drives, floppies, CD, tapes, ...
- Mass-Storage Device Access Methods:
 - Direct Access Storage Devices (DASDs)
 - Hard-drives, floppy-disks, CD-ROMs, ...
 - Sequential Access Storage Devices (SASDs)
 - Tapes (for example, used as backup devices)

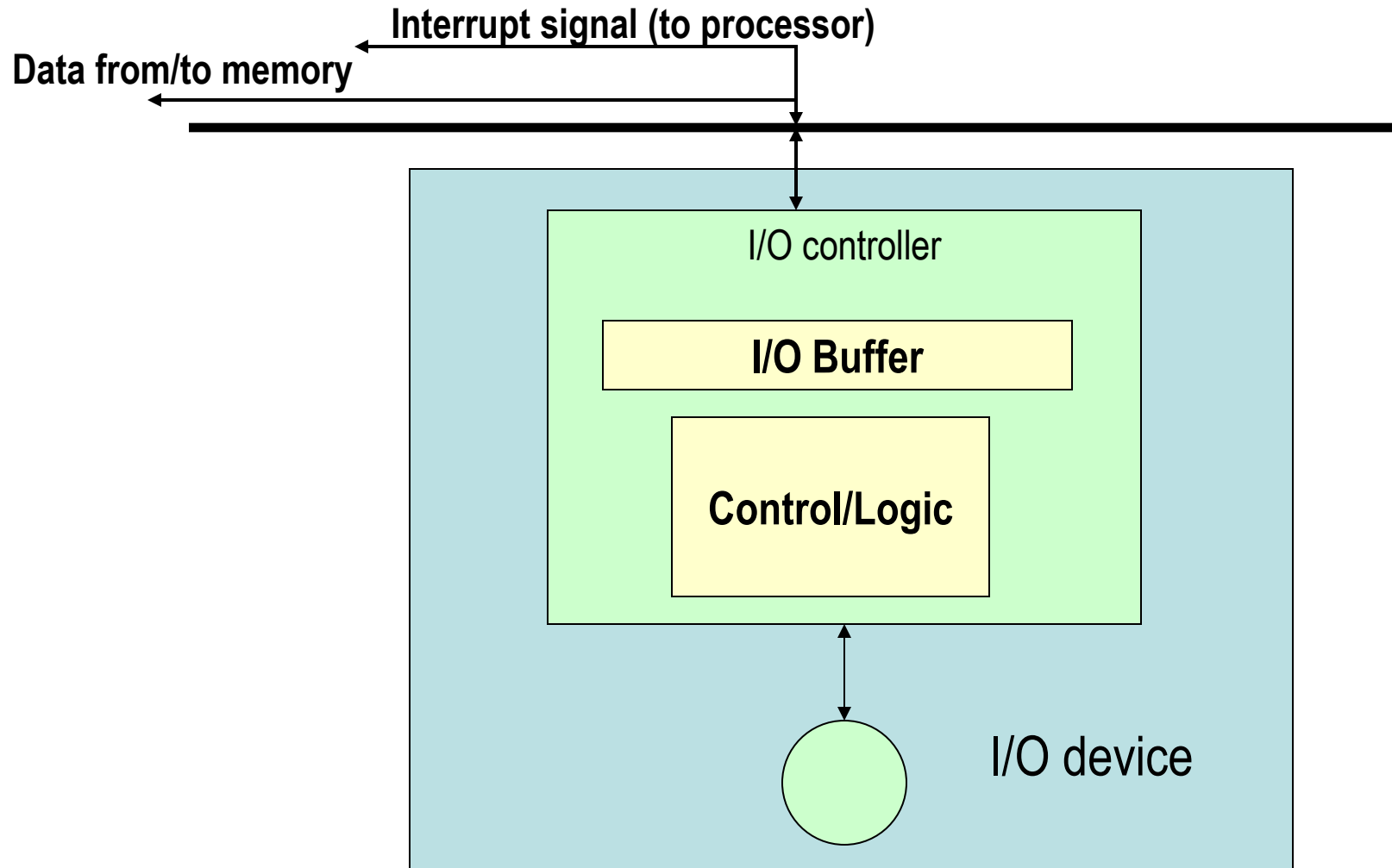


I/O Controllers

- Speed of I/O devices is slow compared to RAM
 - RAM ~ 50 nsec
 - Hard-Drive ~ 10msec = (10,000,000 nsec)
- Solution:
 - I/O Controller, a special purpose processor balance the gap between the speed of processor and I/O device.
 - Data transferred between RAM and memory buffer.
 - Processor free to do something else while I/O controller reads/writes data from/to device into I/O buffer.



Structure of the I/O Subsystem



The ALU Subsystem

- The ALU (Arithmetic/Logic Unit) performs
 - mathematical operations (+, -, x, /, ...)
 - logic operations (=, <, >, and, or, not, ...)
- In today's computers integrated into the CPU
- Consists of:
 - Circuits to do the arithmetic/logic operations.
 - Registers (fast storage units) to store intermediate computational results.
 - Bus that connects the two.



The Control Unit

- Program is stored in memory
 - as machine language instructions, in binary
- The task of the control unit is to execute programs by repeatedly:
 - Fetch from memory the next instruction to be executed.
 - Decode it, that is, determine what is to be done.
 - Execute it by issuing the appropriate signals to the ALU, memory, and I/O subsystems.
 - Continues until the HALT(end) instruction

